

SQL Logic Documentation

Task 1: Average Transaction Value (ATV)

Objective

Calculate the average revenue generated per transaction.

Formula

$ATV = \text{Total Revenue} / \text{Total Transactions}$

SQL Logic

- SUM(Revenue) is used to calculate the total revenue.
- COUNT(TransactionID) is used to count the total number of transactions.
- The total revenue is divided by the total transactions to get the Average Transaction Value.

Business Purpose

ATV helps measure how much customers spend on average in each transaction.

Task 2: Sell-Through Rate

Objective

Measure how efficiently inventory is sold.

Formula

$\text{Sell-Through Rate} = (\text{Units Sold} / \text{Inventory Units}) \times 100$

SQL Logic

- UnitsSold represents the number of products sold.
- InventoryUnits represents the available inventory.
- The formula calculates the percentage of inventory sold.

Business Purpose

A higher sell-through rate indicates strong product demand and efficient inventory management.

Task 3: Category Sell-Through Rate

Objective

Calculate sell-through performance for each product category.

Formula

Category Sell-Through Rate = $(\text{Total Units Sold} / \text{Total Inventory Units}) \times 100$

SQL Logic

- Product_Dimension and Inventory_Dimension are joined using ProductID.
- Data is grouped by Category.
- SUM(UnitsSold) and SUM(InventoryUnits) are calculated for each category.
- The sell-through percentage is calculated for every category.

Business Purpose

This helps identify which product categories are performing well and which categories need inventory optimization.